Database Technologies – Assignment 10

PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
d6_96728_shaurya> DELIMITER $$
d6_96728_shaurya> CREATE PROCEDURE sp_area_calc(IN L INT , IN B INT , OUT area INT, OUT perimeter INT)
    -> BEGIN
    ->     SET area = L*B;
    ->     SET perimeter = 2*(L+B);
    -> END;
    ->
    -> $$
Query OK, 0 rows affected (0.100 sec)

d6_96728_shaurya> DELIMITER ;
d6_96728_shaurya> CALL sp_area_calc(5,6, @area, @perimeter);
Query OK, 0 rows affected (0.015 sec)

d6_96728_shaurya>
d6_96728_shaurya> SELECT @area , @perimeter;
+-------+------------+
| @area | @perimeter |
+-------+------------+
|    30 |         22 |
+-------+------------+
1 row in set (0.002 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
d6_96728_shaurya>
d6_96728_shaurya>
d6_96728_shaurya> DELIMITER $$
d6_96728_shaurya> CREATE PROCEDURE sp_dec(IN num INT, OUT var1 INT , OUT Square_1 INT, OUT CUBE_1 INT)
    -> BEGIN
    ->     SET var1 = num;
    ->     SET Square_1 = num*num;
    ->     SET CUBE_1 = num*num*num;
    ->     INSERT INTO temp VALUES(var1, SQUARE_1, CUBE_1);
    -> END;
    ->
    -> $$
Query OK, 0 rows affected (0.079 sec)

d6_96728_shaurya> DELIMITER ;
d6_96728_shaurya> CALL sp_dec(4, @var1, @Square_1, @CUBE_1);
Query OK, 1 row affected (0.089 sec)

d6_96728_shaurya> SELECT @var1, @Square_1, @CUBE_1;
+-------+-----------+---------+
| @var1 | @Square_1 | @CUBE_1 |
+-------+-----------+---------+
|     4 |        16 |      64 |
+-------+-----------+---------+
1 row in set (0.008 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
d6_96728_shaurya> DELIMITER $$
d6_96728_shaurya> CREATE PROCEDURE sp_temp(IN x FLOAT, IN Flag CHAR, OUT TEMPE FLOAT)
    -> BEGIN
    ->     IF Flag = 'C' THEN
    ->     SET  TEMPE = (9/5)*x + 32;
    ->     ELSE
    ->     SET TEMPE = (x-32)*(5.0/9);
    ->     END IF;
    -> END;
    ->
    -> $$
Query OK, 0 rows affected (0.079 sec)

d6_96728_shaurya> DELIMITER ;
d6_96728_shaurya> CALL sp_temp(43,'C', @TEMPE);
Query OK, 0 rows affected (0.005 sec)

d6_96728_shaurya> SELECT @TEMPE;
+--------------------+
| @TEMPE             |
+--------------------+
| 109.4000015258789 |
+--------------------+
1 row in set (0.005 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d6_96728_shaurya> DELIMITER $$
d6_96728_shaurya>
d6_96728_shaurya> CREATE PROCEDURE sp_inches(
    ->     IN total_inches INT,
    ->     OUT yards INT,
    ->     OUT feet INT,
    ->     OUT inches INT
    -> )
    -> BEGIN
    ->     SET yards = total_inches DIV 36;
    ->     SET feet = (total_inches MOD 36) DIV 12;
    ->     SET inches = total_inches MOD 12;
    -> END $$
Query OK, 0 rows affected (0.106 sec)

d6_96728_shaurya>
d6_96728_shaurya> DELIMITER ;
d6_96728_shaurya> CALL sp_inches(124, @yards, @feet, @inches);
Query OK, 0 rows affected (0.010 sec)

d6_96728_shaurya> SELECT @yards, @feet, @inches;
+--------+-------+---------+
| @yards | @feet | @inches |
+--------+-------+---------+
|      3 |     1 |       4 |
+--------+-------+---------+
1 row in set (0.004 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100

```
d6_96728_shaurya> DELIMITER $$
d6_96728_shaurya>
d6_96728_shaurya> CREATE PROCEDURE sp_product(
    ->     IN num1 FLOAT,
    ->     IN num2 FLOAT,
    ->     OUT result VARCHAR(50)
    -> )
    -> BEGIN
    ->     IF num1 * num2 >= 100 THEN
    ->         SET result = 'Product is >= 100';
    ->     ELSE
    ->         SET result = 'Product is < 100';
    ->     END IF;
    -> END $$
Query OK, 0 rows affected (0.100 sec)

d6_96728_shaurya>
d6_96728_shaurya> DELIMITER ;
d6_96728_shaurya> CALL sp_product(10, 12, @res);
Query OK, 0 rows affected (0.012 sec)

d6_96728_shaurya> SELECT @res;
+-------------------+
| @res              |
+-------------------+
| Product is >= 100 |
+-------------------+
1 row in set (0.004 sec)
```